

SYLLABUS

Of

3 YEARS BCA COURSE

Under



PATLIPUTRA UNIVERSITY
PATNA

From
ACADEMIC SESSION 2025-2026

Minimum Eligibility for Admission:

**10+2 / Intermediate or Equivalent in any Discipline with minimum
45% Marks from a Recognised Board.**

(Himanshu Ojha)
Dept. of MCA
College of Commerce
Arts & Science, Patna

(Aditya Singh Suman)
Dept of MCA
A N College, Patna

(Mrs. Anupam Singh)
Dept of MCA & BCA
Arcade Business College
Patna

BCA – 1st YEAR

HONS. PAPER	SUBJECTS	THEORY MARKS	INTERNAL PRACTICAL MARKS
I	Computer Fundamentals	75	25
	Office Automation Application (MS Office)		
	Web Application – HTML, CSS & JavaScript		
II	Programming Logic and Algorithm with C Language	75	25
	Operating System with Linux and Windows		
	Software Analysis and Design (SAD)		
SUBSIDIARY PAPERS	SUBJECTS	MARKS	
I	Subsidiary I	100	
II	Subsidiary II	100	
Composition	Hindi	100	

BCA – 2nd YEAR

HONS. PAPER	SUBJECTS	THEORY MARKS	INTERNAL PRACTICAL MARKS
III	Data Structure using C	75	25
	Digital System And Computer Organization & Architecture		
	Object Oriented Technology using C++		
IV	Core Java	75	25
	Computer Network and Internet Technology		
	Python with Mini Project		
SUBSIDIARY PAPERS	SUBJECTS	MARKS	
I	Subsidiary I	100	
II	Subsidiary II	100	
Composition	Hindi	100	

Hons Paper. III:

Full Marks – 100

(Theory – 75, Practical – 25)

1.) Data Structure using C

Introduction to data structure

Concept of data structure

Abstract data structure

Analysis of algorithm

The Concept of list

Stack and Queue

Introduction to stack and primitive operation on stack

Stack as an abstract data structure

Multiple Stack

Stack Application : infix, postfix, prefix and recursion

Introduction to queue and primitive operation on queue

Queues as an abstract data structure

Circular queue

Dequeue

Priority queue

Linked List

Introduction to linked list of stack

The linked list of queue

Header nodes

Doubly Linked list

Circular Linked List

Stack and Queue as a circular linked list

Application of linked list

Tree

Basic Terminology

Binary Tree

The tree representation as array and linked list

Binary Tree representation

Traversal of binary tree: Inorder, Preorder, Postorder

Application of binary tree

Threaded binary tree

B tree and Height Balanced tree, representation of B+ and B*

Counting Binary trees

Searching and Sorting

Sequential searching

Binary Search

Selection Sort

Insertion sort

Quick sort

Bubble sort

Radix sort

Heap sort

Comparison of sorting methods

Tables and Graphs

Hash table

Collision resolution technique

Introduction to graph

Definition

Terminology

Directed, Undirected, & Weighted graph

Representation of graphs

Graph Traversal Depth first & breath first search

Spanning Tree, minimum Spanning tree

The Basic Greedy Strategy for Kruskal and Prims.

2.) Digital System and Computer Organization & Architecture

Introduction, Data Representation ,Number Systems, Decimal Representation in Computers, Alphanumeric Representation, Data Representation For Computation, Fixed Point Representation, Decimal Fixed Point Representation, Floating Point Representation.

Logic Gates, Logic Circuits ,Combinational Circuits, Canonical and Standard Forms, Minimization of Gates, Design of Combinational Circuits, Adders ,Encoder ,Decoders ,Multiplexer, RAM, Registers

Sequential Circuits, Flip Flops , Types of Flip- Flops

K-Map

Error Detection And Correction Codes

Basic Computer Organization: Introduction to Microprocessor, Block Diagram of computer, Selection criteria of microprocessor, Instruction, Types of Instruction, instruction Cycle, Register Organization, Bus, Interrupts, Interrupt Cycle, Booting process.

Memory Organization: Types of memory, Cache Memory, memory Mapping, Virtual Memory Concept, DMA.

I/O Organization: I/O Modules, Mode of data transfer from I/O to memory. Introduction to pipelining.

3.) Object Oriented Technology using C++

- Introduction to object Oriented Programming(OOP) and c++
- Objects
- Polymorphisms
- Inheritance
- C++ fundamental
- Classes and Objects
- Function and Overloading

- Operator Overloading
- Constructor and destructor
- Multiple Inheritance
- Passing Objects to function
- Array of objects
- Pointers to Objects
- C++ I/O Class library
- C++ Stream
- C++ Pre-defined streams
- C++ Stream Classes

1.) Core Java

Java program structure, tokens, Java Virtual Machine, Constants and variables, Data types, Declaration of variables and scope of variable, type casting

Operators : Arithmetic, relational, logical, assignment, increment and decrement, conditional, Bit wise, Special Expression and its evaluation.

Decision making and branching: if statement, if else statement, nesting of if else statement, if else ladder, switch statement:operatorloops: do-while, for, while, jumping loops, labeled loops

Classes, Objects, Methods: Defining class, adding variable and methods, creating object, accessing class member, constructor, methods of overloading, static member, nesting of methods

Inheritance: Extending, a class, Overriding methods, final variables and methods, final classes, finalize methods, abstracts methods and classes, visibility controls

Arrays: One dimensional and two dimensional, string and vectors, wrapper class

Interface and package: Defining interface, Extending interfaces, Implementing interfaces, Accessing Interface, Creating and importing packages

Multithreading : Creating Threads, Extending thread class, stopping and blocking thread, life cycle of thread, thread exception, thread priority, synchronization, implementing the run ableinterface.

Applet Programming: Local and Remote Applet vs. Application writing Applet, Applet life cycle, Creating and Executing Applets, designing a web page, Applet tag, adding Applet to HTML, Running the Applets, Passing parameters to Applets, Aligning the display

2.) Computer Network and Internet Technology

Introduction to the N/W. Basic Requirements for network.Network Topologies.Classification of Computer Network.Data and Signal. Communication Channels: Twisted Wide, Co-axial

Wire, optical fibre, Radiowave Communication, Microwave Communication, Satellite Communication, VSAT.

Switching Techniques: Circuit Switching, Message Switching, Packet Switching, FEP, Repeater, Bridge, Gateway, Token Passing CSMA/CD protocols.

Network Models: ISO: OSI model, TCP/IP Model

Network devices: Hub, Switch, Bridge, Router, Gateways, Modem

Internet Technology:

Internet, Evolution, Protocols, Interface Concept, Internet Vs Intranet, Growth of internet, DNS, ISP, Connectivity Dial Up, Leased Line, VSAT ect., URLs, Domain names, Portals, application.

E-mail Concept, POP and Web Based E-mail, address, Basics of sending and receiving E-mail, protocols, mailing list, Free E-mail services.

3.) Python with Mini Projects

An Introduction to Python

Python Syntax, Python Installation, The print statement, Comments, Python Data Structures & Data Types, String Operations, Simple Input & Output, Simple Output Formatting, Operators in python

Python Program Flow: Indentation, If statement, loop statements, Break & Continue, Assert

Functions & Modules: Functions Parameters, Variable Arguments, Scope of a Function, Function Documentations, Lambda Functions & map

Create a Module: Standard Modules

Exceptions Handling

File Handling: File handling Modes, Reading Files,

OOPs In Python